

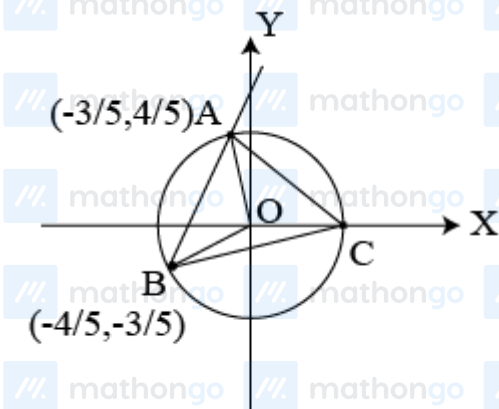
Hints & Solutions

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Q1 (3) - Single Correct

Solving $y = 7x + 5$ and the circle

$$x^2 + y^2 = 1$$



$$A\left(-\frac{3}{5}, \frac{4}{5}\right) \text{ and } B\left(-\frac{4}{5}, -\frac{3}{5}\right)$$

$$m_{OA} = -\frac{4}{3}, m_{OB} = \frac{3}{4}$$

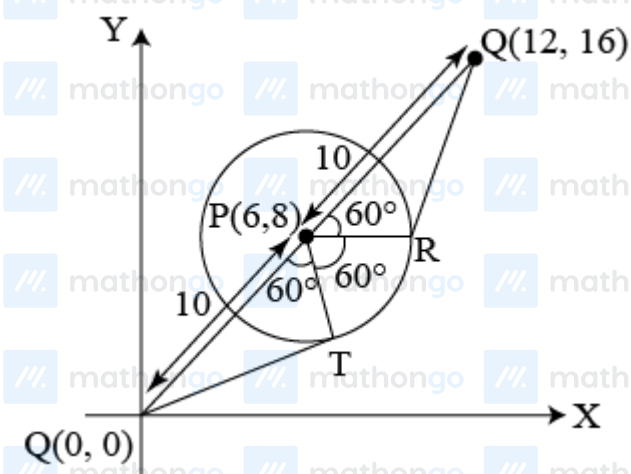
Hence, $\angle AOB = 90^\circ$

$$\Rightarrow \angle ACB = \frac{\pi}{4} = \tan^{-1}(1)$$

Q2 (4) - Single Correct

Let $O(0,0)$, $P(6,8)$ and $Q(12,6)$

As shown in the figure, the shortest route consists of tangent OT , minor arc TR and tangent RQ .



Since, $OP = 10$, $PT = 5$ and $\angle OPT = 90^\circ$, it follows that $\angle OPT = 60^\circ$ and $OT = 5\sqrt{3}$

By similar reasoning,

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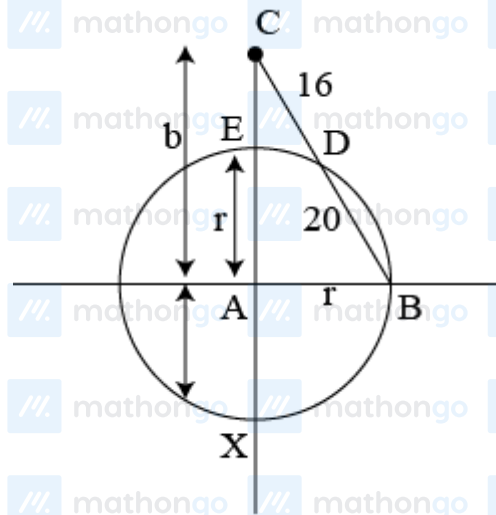
$$\angle QPR = 60^\circ \text{ and } QR = 5\sqrt{3}$$

Because O, P and Q are collinear.

$\angle RPT = 60^\circ$, so arc TR is of length $5\frac{\pi}{3}$

Hence, the length of the shortest route is $\left(10\sqrt{3} + \frac{5\pi}{3}\right)$.

Q3 (2) - Single Correct



From figure $b^2 + r^2 = (36)^2 \dots(i)$

Also, $CD \cdot CB = CE \cdot CX$

$$\Rightarrow 16 \cdot (36) = (b - r)(b + r) \dots(ii)$$

From Eqs. (i) and (ii), we get

$$2b^2 = 36(36 + 16) = 36 \cdot (52)$$

$$\Rightarrow b = 6\sqrt{26} \text{ units}$$

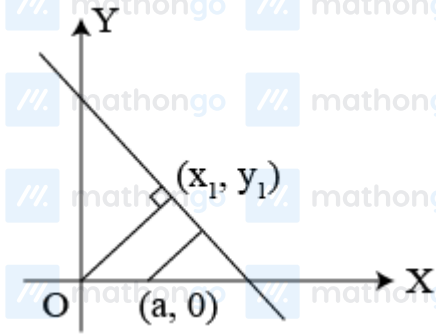
Q4 (1) - Single Correct

Let the equation of the variable line is

$$xx_1 + yy_1 - (x_1^2 + y_1^2) = 0$$

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$$P_1 P_2 = \left| \frac{x_1^2 + y_1^2}{\sqrt{x_1^2 + y_1^2}} \right| \left| \frac{ax_1 - (x_1^2 + y_1^2)}{\sqrt{x_1^2 + y_1^2}} \right| = k^2$$

$$\text{i.e. } |x_1^2 + y_1^2 - ax_1| = k^2$$

$$\text{Locus is } x^2 + y^2 - ax \pm k^2 = 0$$

$$\therefore r^2 = \frac{a^2}{4} - (\pm k^2)$$

$$\Rightarrow r^2 = \frac{a^2}{4} - k^2 \quad [\text{not possible}]$$

$$\text{Thus, } r^2 = \frac{a^2}{4} + k^2$$

Q5 (1) - Single Correct

Equation of circle is $x^2 + y^2 = 1$ and

equation of line passing through $(x_1, -1)$ and $(x_2, -1)$ is

$$y + 1 = \frac{2}{x_2 - x_1}(x - x_1)$$

$$\Rightarrow (x_2 - x_1)y + x_2 - x_1 = 2x - 2x_1$$

$$\Rightarrow 2x - (x_2 - x_1)y - (x_1 + x_2) = 0 \dots (i)$$

Since, the above line is tangent to circle.

$\therefore$ Distance of line from centre of circle = Radius

$$\Rightarrow \left| \frac{-(x_1 + x_2)}{\sqrt{4 + (x_2 - x_1)^2}} \right| = 1$$

$$\Rightarrow (x_1 + x_2)^2 = 4 + (x_2 - x_1)^2$$

$$\Rightarrow x_1^2 + x_2^2 + 2x_1x_2 = 4 + x_2^2 + x_1^2 - 2x_1x_2$$

$$\Rightarrow 4x_1x_2 = 4$$

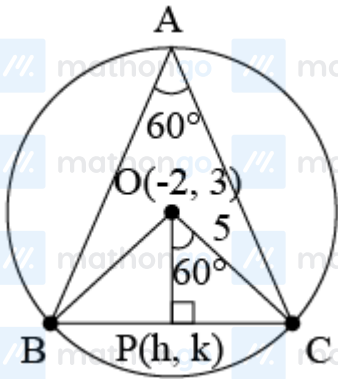
$$\therefore x_1x_2 = 1$$

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Q6 (2) - Single Correct

In $\triangle OPC$



$$\frac{OP}{5} = \cos 60^\circ \Rightarrow OP = 2.5$$

$$\Rightarrow \sqrt{(h+2)^2 + (k-3)^2} = 2.5$$

$$\text{or } (x+2)^2 + (y-3)^2 = 6.25$$

Q7 (3) - Single Correct

Let the equation of circle be

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Since, $\left(a, \frac{1}{a}\right), \left(b, \frac{1}{b}\right), \left(c, \frac{1}{c}\right)$

and $\left(d, \frac{1}{d}\right)$ lie on the circle of radius 4.

$$\therefore a^2 + \frac{1}{a^2} + 2ga + \frac{2f}{a} + c = 0$$

$$\Rightarrow a^4 + 2ga^3 + ca^2 + 2fa + 1 = 0$$

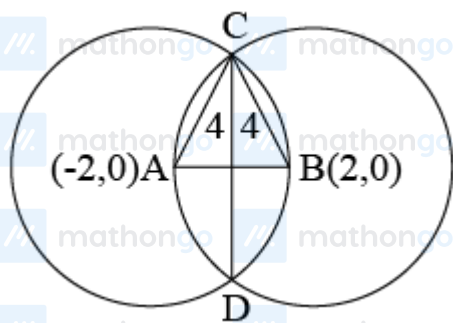
$$\Rightarrow abcd = 1$$

Q8 (1) - Single Correct

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Circles with centres $(2,0)$ and $(-2,0)$ each with radius 4 units is shown below.



$\Rightarrow$ Y-axis is their common chord.

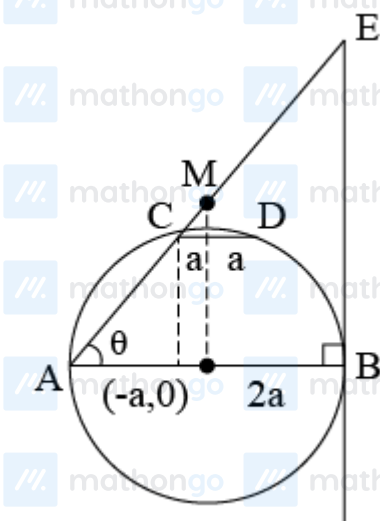
$\therefore \triangle ABC$ is equilateral.

Hence, area of $ADBC = \frac{2\sqrt{3}}{4}(4)^2 = 8\sqrt{3}$ sq units

So, the correct option is (1).

Q9 (4) - Single Correct

Coordinates of $C(-a, a\sqrt{3})$ on $x^2 + y^2 = 4a^2$.



$$\therefore \tan \theta = \left| \frac{a\sqrt{3}}{a} \right| = \sqrt{3}$$

$$\Rightarrow \theta = 60^\circ$$

Now, in $\triangle ABE$, $\cos 60^\circ = \frac{4a}{AE}$

$$\Rightarrow AE = 8a = 2(AB)$$

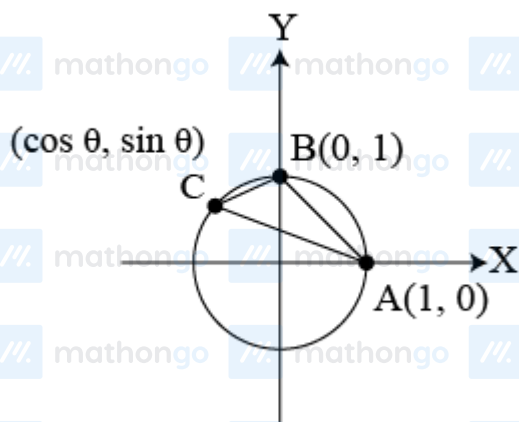
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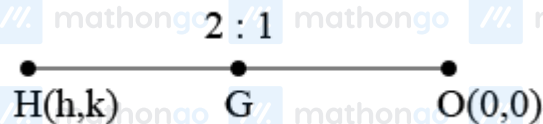
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Q10 (1) - Single Correct

Let $C(\cos \theta, \sin \theta)$



$H(h, k)$ be the orthocentre of the ΔABC



$$\left(\frac{\cos \theta + 1}{3}, \frac{\sin \theta + 1}{3} \right)$$

$$\Rightarrow h = 1 + \cos \theta$$

$$\text{and } k = 1 + \sin \theta$$

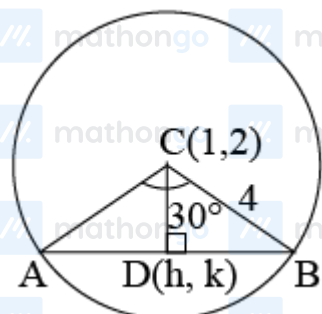
$$\therefore (h - 1)^2 + (k - 1)^2 = 1$$

$$\text{or } (x - 1)^2 + (y - 1)^2 = 1$$

$$\Rightarrow x^2 + y^2 - 2x - 2y + 1 = 0$$

Q11 (3) - Single Correct

In ΔCDB



$$\frac{CD}{4} = \cos 30^\circ$$

$$\Rightarrow \sqrt{(h - 1)^2 + (k - 2)^2} = 4 \cdot \frac{\sqrt{3}}{2}$$

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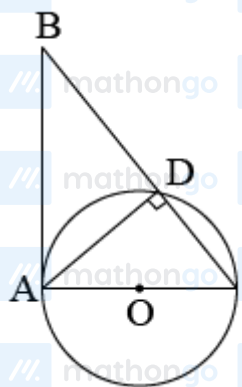
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or locus is, $(x-1)^2 + (y-2)^2 = 12$

or $x^2 + y^2 - 2x - 4y - 7 = 0$

Q12 (4) - Single Correct

Since, AC is the diameter, so $\angle ADC = 90^\circ$



Now, in $\triangle ABC$, $\triangle DAC$

$$\Rightarrow \angle A = \angle D = 90^\circ$$

$$\angle C = \angle C = \text{Common}$$

Hence, by AA similarity, $\triangle ABC \sim \triangle DAC$.

$$\text{So, } \frac{AB}{AD} = \frac{BC}{AC} = \frac{AC}{CD}$$

$$AC = \frac{AD \times BC}{AB}$$

$$\Rightarrow AC = \frac{AD \cdot \sqrt{AB^2 + AC^2}}{AB}$$

$$\Rightarrow AC \cdot AB = AD \sqrt{AB^2 + AC^2}$$

$$\Rightarrow AC^2 \cdot AB^2 = AD^2 \cdot AB^2 + AD^2 \cdot AC^2$$

$$\Rightarrow AC^2 [AB^2 - AD^2] = AD^2 AB^2$$

$$\therefore AC = \frac{AD \cdot AB}{\sqrt{AB^2 - AD^2}}$$

Q13 (1) - Single Correct

Let $A(a, b)$ and $G(h, k)$.

Now, A, G, O are collinear.

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$$\Rightarrow h = \frac{a}{3}$$

$$\Rightarrow a = 3h$$

and similarly $b = 3k$.

Now, (a, b) lies on the circle $x^2 + y^2 = 9$

$$\therefore a^2 + b^2 = 9$$

$$\text{Or } (3h)^2 + (3k)^2 = 9$$

$$\Rightarrow x^2 + y^2 = 1$$

Q14 (1,2) - Multiple Correct

Line $y = 1 + c(x + 3)$ is tangent to circle $x^2 + y^2 = 1$

$$\therefore p = r$$

$$\Rightarrow \left| \frac{1+3c}{\sqrt{1+c^2}} \right| = 1$$

$$\Rightarrow (1+3c)^2 = 1 + c^2$$

$$\text{or } 9c^2 + 6c + 1 = c^2 + 1$$

$$\Rightarrow 8c^2 + 6c = 0$$

$$\therefore c = 0, -3/4$$

Q15 (2,4) - Multiple Correct

Note that, Δ is right angled at $(0,6)$. Centre of the circle is $(2,3)$. Slope of the line joining origin to the centre is $3/2$. Take parametric equation of a line through $(2,3)$ with

$$\tan \theta = -2/3 \text{ as } \frac{x-2}{\cos \theta} = \frac{y-3}{\sin \theta} = \pm r$$

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where, $r = \sqrt{13}$

$$\therefore x = 2 \pm r \cos \theta$$

$$\text{and } y = 3 \pm r \sin \theta$$

$$\text{Also, } x = 2 \pm \sqrt{13} \cdot \frac{3}{\sqrt{13}}$$

$$\text{and } y = 3 \pm \sqrt{13} \cdot \frac{2}{\sqrt{13}}$$

$$\text{or } (x = 5, y = 1) \text{ and } (x = -1, y = 5)$$

Required points are (5,1) and (-1,5)

Q16 (1,2,4) - Multiple Correct

$$S_1 : x^2 + y^2 = 4$$

$$\text{and } S_2 : x^2 + y^2 - 2x - 4y + 4 = 0$$

Centre (0, 0), radius = 2 and centre (1, 2), radius = 1

$$(1) d = \text{Distance between centres} = \sqrt{5} \text{ units}$$

$$r_1 + r_2 = 3$$

$$\Rightarrow |r_1 - r_2| = 1$$

$$\therefore |r_1 - r_2| < d < r_1 + r_2$$

$\Rightarrow$ These are intersecting.

Hence, number of tangents = 2

(2) $P(h, k)$ power of point P is same w.r.t. these two circles.

$$\therefore \sqrt{h^2 + k^2 - 4} = \sqrt{h^2 + k^2 - 2h - 4k + 4}$$

$$\Rightarrow -4 = -2h - 4k + 4$$

$$\text{or } x + 2y - 4 = 0$$

$$(3) y\text{-intercept of } S_1 \text{ is } 2\sqrt{4} = 4$$

$$y\text{-intercept of } S_2 \text{ is } 2\sqrt{4 - 4} = 0$$

$$\therefore \text{Sum of } y\text{-intercept} = 4$$

$$(4) 2(0 + 0) = -4 + 4$$

$$\text{i.e. } 2(g_1g_2 + f_1f_2) = C_1 + C_2$$

$\therefore$ Circles are orthogonal.

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Q17 (1,3) - Multiple Correct

$$(d^2 - b) l^2 + 2dl + 1 = bm^2$$

$$\Rightarrow d^2 l^2 + 2dl + 1 = b(l^2 + m^2)$$

$$\Rightarrow \left| \frac{dl+1}{\sqrt{l^2+m^2}} \right| = (\sqrt{b})^2$$

$$\Rightarrow \text{Centre } (d, 0) \text{ and radius } = \sqrt{b} \Rightarrow (x-d)^2 + (y-0)^2 = b$$

Q18 (2,3) - Multiple Correct

The equation of circle

$$x^2 + y^2 - 8x - 16y + 60 = 0 \dots(i)$$

Equation of chord of contact from $(-2,0)$ is

$$-2x - 4(x-2) - 8y + 60 = 0$$

$$3x + 4y - 34 = 0 \dots(ii)$$

$$x^2 + \left(\frac{34-3x}{4}\right)^2 - 8x - 16\left(\frac{34-3x}{4}\right) + 60 = 0$$

$$\Rightarrow 16x^2 + 1156 - 204x + 9x^2 - 128x - 2176 + 192x + 960 = 0$$

$$\Rightarrow 5x^2 - 28x - 12 = 0$$

$$\Rightarrow (x-6)(5x+2) = 0$$

$$\therefore x = 6, -\frac{2}{5}$$

$$\text{So, the points are } (6, 4), \left(-\frac{2}{5}, \frac{44}{5}\right)$$

Q19 (1,3,4) - Multiple Correct

Case I If lines form a Δ , then $n = 4$

i.e. 3 excircles and 1 incircle.

Case II If lines are concurrent

or all 3 parallel, then $n = 0$

Case III If two are parallel

and third cuts, then $n = 2$

Q20 (2,3,4) - Multiple Correct

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Radical centre is $(1, -1)$ Radius of the circle orthogonally to S_1, S_2, S_3 , is 1 and has the equation

$$x^2 + y^2 - 2x + 2y + 1 = 0$$

Q21 (1,2,4) - Multiple Correct

Since, $(a, 0)$ is a point on the diameter along X -axis.

$$\therefore -2 \leq a \leq 2 \Rightarrow 0 \leq a^2 \leq 4$$

Let $f(x) = x^2 - 4x - a^2$. Clearly, discriminant of $f(x) = 0$ is positive.

So, it has real and distinct roots.

Now

$$f(0)f(-1) = (-a^2)(5 - a^2) < 0 \quad (\because 0 \leq a^2 \leq 4)$$

$$\Rightarrow f(x) = 0 \text{ has root in } (-1, 0]$$

$$\text{Also, } f(2)f(5) = (-4 - a^2)(5 - a^2) < 0$$

$$\Rightarrow f(x) = 0 \text{ has a root in } [2, 5]$$

Q22 (1,3) - Multiple Correct

Let tangents intersect at (α, β) , then the line joining the two points at which tangents are drawn is

(i) common chord of C_1 and C_2 is

$$C_1 - C_2 = 0$$

$$\Rightarrow 2x + 4y - 18 = 0$$

$$\text{or } x + 2y - 9 = 0$$

(ii) chord of contact when tangents are drawn from (α, β) to C_1

$$\Rightarrow 2x + 4y - 18 = 0$$

$$\text{and } \alpha x + \beta y - 25 = 0$$

must be identical on comparing, we get

$$\frac{\alpha}{2} = \frac{\beta}{4} = \frac{25}{18}$$

$$\therefore \text{Tangents intersect at } \left(\frac{25}{9}, \frac{50}{9}\right)$$

Q23 (2,3) - Multiple Correct

The centre and radius of the given circle are $\left(\frac{1}{2}, \frac{-3}{2}\right)$ and $\sqrt{\frac{5}{2}}$, respectively.

Let $y = mx$ be the equation of L_1 Then, p_1 = Length of the intercept on L_1

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$$\Rightarrow p_1 = 2\sqrt{\left(\sqrt{\frac{5}{2}}\right)^2 - \left(\frac{m+3}{2\sqrt{m^2+1}}\right)^2}$$

$$\Rightarrow p_1 = 2\sqrt{\frac{5}{2} - \frac{(m+3)^2}{4(m^2+1)}}$$

and $p_2 =$ Length of the intercept on L_2

$$\Rightarrow p_2 = 2\sqrt{\left(\sqrt{\frac{5}{2}}\right)^2 - (\sqrt{2})^2}$$

$$\Rightarrow p_2 = 2\sqrt{\frac{5}{2} - 2} = \sqrt{2}$$

Now, $p_1 = p_2$

$$2\sqrt{\frac{5}{2} - \frac{(m+3)^2}{4(m^2+1)}} = \sqrt{2}$$

$$\Rightarrow 5 - \frac{(m+3)^2}{4(m^2+1)} = 1$$

$$\Rightarrow 7m^2 - 6m - 1 = 0$$

$$\Rightarrow (m-1)(7m+1) = 0$$

$$\Rightarrow m = 1, -\frac{1}{7}$$

So, the equation of line L_1 are

$$y - x = 0$$

$$\Rightarrow x - y = 0$$

$$7y = -x$$

$$\Rightarrow x + 7y = 0$$

Q24 (1,2,3,4) - Multiple Correct

$x^2 + y^2 - 2x + 4y = 0$ has centre $C(1, -2)$

Distance of $C(1, -2)$ from $3x - y = 0$

$$\Rightarrow \frac{|3+2|}{\sqrt{9+1}} = \frac{5}{\sqrt{10}} = \sqrt{\frac{5}{2}}$$

Distance of $C(1, -2)$ from $x + 3y = 0$

$$\Rightarrow \frac{|1-6|}{\sqrt{9+1}} = \sqrt{\frac{5}{2}}$$

Distance of $C(1, -2)$ from $x + 3y + 10 = 0$

$$\Rightarrow \frac{|1-6+10|}{\sqrt{1+9}} = \sqrt{\frac{5}{2}}$$

Distance of $C(1, -2)$ from $3x - y - 10 = 0$

$$\Rightarrow \frac{|3+2-10|}{\sqrt{9+1}} = \sqrt{\frac{5}{2}}$$

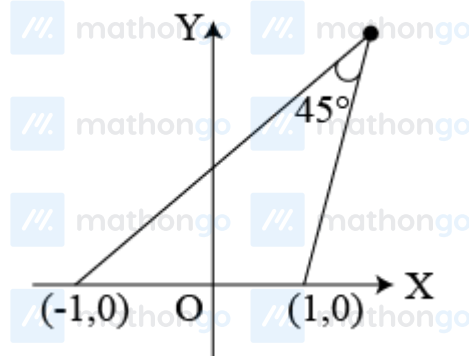
Since, chords equidistance from the centre are equal.

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Q25 (3,4) - Multiple Correct



Here, $m_1 = \frac{k}{h-1}$ and $m_2 = \frac{k}{h+1}$

$$\therefore \tan 45^\circ = \left| \frac{\frac{k}{h-1} - \frac{k}{h+1}}{1 + \frac{k^2}{h^2-1}} \right|$$

$$= \left| \frac{k(h+1-h+1)}{h^2-1+k^2} \right|$$

$$\therefore h^2 + k^2 - 1 = \pm 2k$$

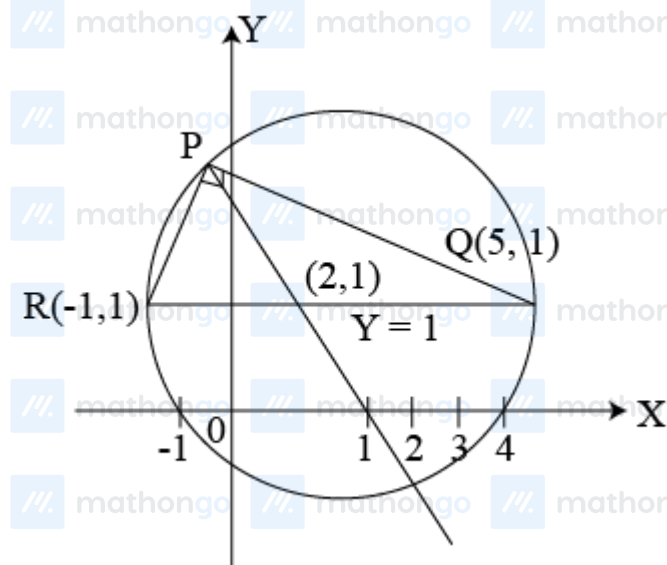
Locus is, $x^2 + y^2 \pm 2y - 1 = 0$

$$\therefore x^2 + (y \pm 1)^2 = (\sqrt{2})^2$$

$\Rightarrow$ Centre (0,1) or (0,-1) and radius $= \sqrt{2}$.

Q26 (2) - Paragraph 1

From figure, $A \cap B \cap C$ is 1.



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Q27 (3) - Paragraph 1

When $y = 1$

$$x^2 - 4x - 5 = 0$$

$$\Rightarrow (x - 5)(x + 1) = 0$$

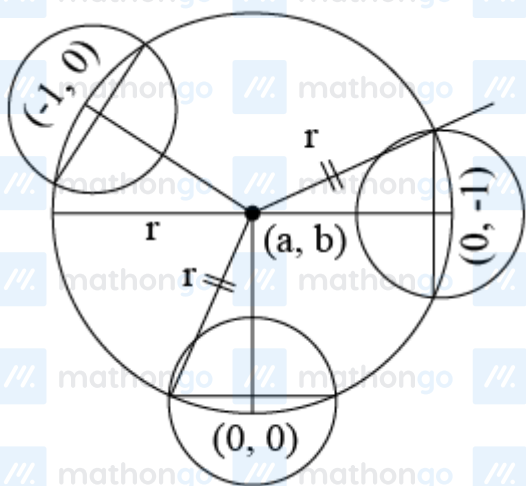
$$\therefore x = -1, 5$$

$$\text{Thus, } (x + 1)^2 + (y - 1)^2 + (x - 5)^2 + (y - 1)^2 = QR^2 = 36$$

Q28 (3) - Paragraph 2

$$r^2 = a^2 + b^2 + 1 = (a + 1)^2 + b^2 + 4$$

$$= (a^2) + (b + 1)^2 + 4$$



$$\therefore a^2 + b^2 + 1 = a^2 + 2a + 1 + b^2 + 4$$

$$\Rightarrow a = -2$$

$$\text{and } a^2 + (b + 1)^2 + 4 = (a + 1)^2 + b^2 + 4$$

$$\Rightarrow 2a = 2b \Rightarrow b = -2$$

$$\therefore r^2 = 9 \Rightarrow r = 3 \text{ units}$$

Q29 (4) - Paragraph 2

$$S_1 - S_2 = 0 \Rightarrow x = 1$$

$$S_2 - S_3 = 0 \Rightarrow y = 1$$

$$\therefore \text{Radical centre } (1, 1) \text{ Radius } L_T = \sqrt{S_1} = 1$$

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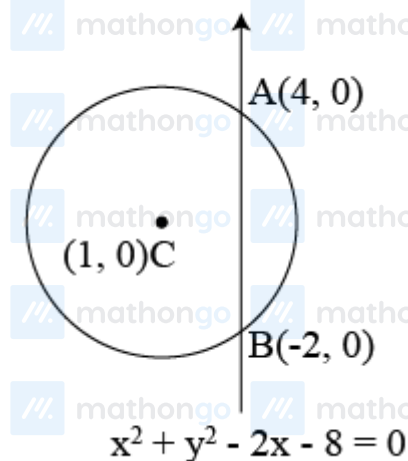
$\therefore$ Equation of circle is $(x - 1)^2 + (y - 1)^2 = 1$

$\Rightarrow$ Radius = 1 and $a = 1, b = 1 \Rightarrow a + b + r = 3$

Q30 (3) - Paragraph 3

Solving $y = 0$

and $x^2 + y^2 - 2x - 8 = 0$



$\Rightarrow x^2 - 2x - 8 = 0$

i.e. $x = 4, -2$

$\therefore A(4, 0)$ and $B(-2, 0)$

$\Rightarrow$ Distance between A and B

= 6 units

Q31 (3) - Paragraph 3

Now, tangent at $A(4, 0)$, is $3x - \lambda y - 12 = 0$ and tangent at $B(-2, 0)$ is $-3x - \lambda y - 6 = 0$

On solving, $x = 1, y = -\frac{9}{\lambda}$

It lies on $x + 2y + 5 = 0$

$\Rightarrow 1 - \frac{18}{\lambda} + 5 = 0$

$\therefore$ Required equation of circle, is

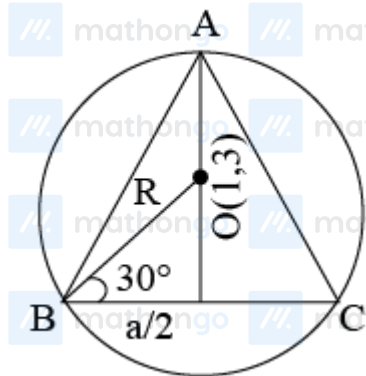
$x^2 + y^2 - 2x - 6y - 8 = 0$

having radius, $R = \sqrt{1 + 9 + 8} = 3\sqrt{2}$

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$$\therefore \frac{a/2}{R} = \cos 30^\circ$$



$$\Rightarrow a = 2R \cdot \frac{\sqrt{3}}{2}$$

$$\Rightarrow a = \sqrt{3}R = 3\sqrt{6}$$

$$\text{Thus, area of equilateral } \Delta = \frac{\sqrt{3}}{4} (3\sqrt{6})^2 = \frac{27\sqrt{3}}{2} \text{ sq units}$$

Q32 (2) - Paragraph 4

As lines are perpendicular

$$\therefore a - 2 = 0$$

$$\Rightarrow a = 2 [\text{coefficient of } x^2 + \text{coefficient of } y^2 = 0]$$

$$\text{Using } \Delta = 0 \Rightarrow c = -3$$

$$[\Delta = abc + 2fgh - af^2 - bg^2 - ch^2]$$

Hence, two lines are

$$x + 2y - 3 = 0 \text{ and } 2x - y + 1 = 0$$

$$x\text{-intercepts, } x_1 = 3, x_2 = \frac{-1}{2}$$

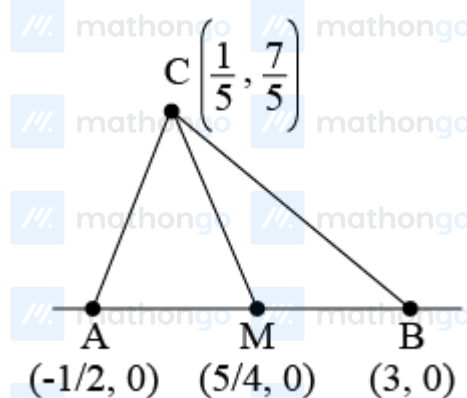
$$y\text{-intercepts, } y_1 = \frac{3}{2}, y_2 = 1$$

$$\Rightarrow x_1 + x_2 + y_1 + y_2 = 5$$

Q33 (3) - Paragraph 4

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$$\begin{aligned}(CM)^2 &= \left(\frac{5}{4} - \frac{1}{5}\right)^2 + \frac{49}{25} \\ &= \frac{49}{16} \\ \therefore CM &= \frac{7}{4} \text{ units}\end{aligned}$$

Q34 (4) - Paragraph 4

Circumcircle of $\triangle ABC$ is $\left(x + \frac{1}{2}\right)(x - 3) + y^2 = 0$

$$\Rightarrow (2x + 1)(x - 3) + 2y^2 = 0$$

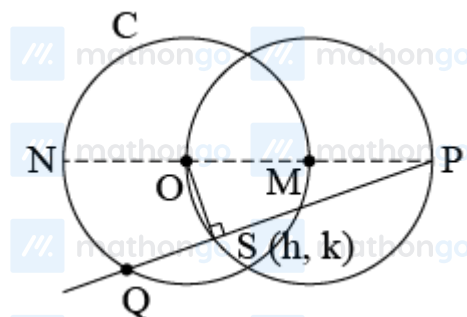
$$\Rightarrow x^2 + y^2 - \frac{5}{2}x - \frac{3}{2} = 0$$

and $x^2 + y^2 - 4y + k = 0$ are orthogonal.

$$\therefore k = \frac{3}{2}$$

Q35 (4) - Paragraph 5

Locus of S is a part of circle with OP as diameter passing inside the circle C .



Q36 (4) - Paragraph 5

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$$(PR) \cdot (PQ) = PT^2$$

$$= (PN)(PM) = (d-r)(d+r) = d^2 - r^2$$

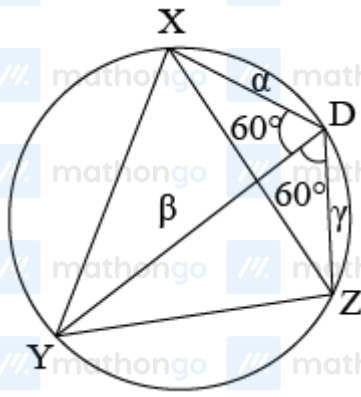
$$= (PS - SR)(PS + SQ) = PS^2 - SQ^2$$

$$= PS^2 - (SQ)(SR)$$

$$\therefore (PQ)(PR) \neq (PS)^2$$

Q37 (1) - Paragraph 5

Using Ptolemy's theorem,



$$\begin{aligned} (YD)(XZ) &= (XY)(ZD) + (YZ)(XD) \\ &= XZ(ZD + XD), \quad (\because XY = YZ = ZX) \end{aligned}$$

$$\beta = \gamma + \alpha$$

Q38 (2) - Integer Type

$$\text{As, } x^2 = y^2$$

$$\Rightarrow y = \pm x$$

$$\text{and } (x - A)^2 + y^2 = 1$$

has centre on X -axis.

Hints & Solutions

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Thus, $AP < r \Rightarrow \frac{|A|}{\sqrt{2}} < 1$

$\Rightarrow |A| < \sqrt{2}$

i.e. $-\sqrt{2} < A < \sqrt{2}$

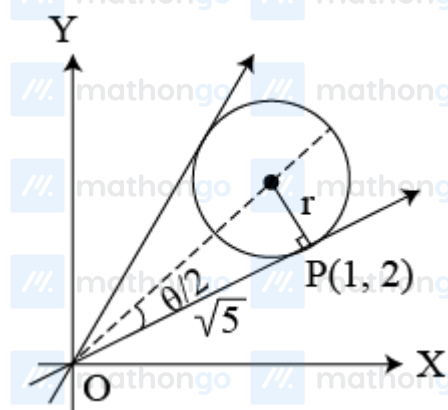
$\Rightarrow A = \{-1, 0, 1\}$ but $A \neq 0$ as only two solutions.

Thus, A can take only 2 values.

Q39 (5) - Integer Type

$y = 2x$ and $y = 3x$ are tangents.

$\tan \theta = \frac{3-2}{1+3(2)} = \frac{1}{7} \dots(i)$



$\therefore \tan \frac{\theta}{2} = \frac{r}{\sqrt{5}}$

$\Rightarrow r = \sqrt{5} \tan \frac{\theta}{2} \dots(ii)$

From Eqs. (i) and (ii), we get

$\tan \theta = \frac{2 \tan \frac{\theta}{2}}{1 - \tan^2 \frac{\theta}{2}}$

$\Rightarrow \frac{1}{7} = \frac{2 \cdot \frac{r}{\sqrt{5}}}{1 - \frac{r^2}{5}}$

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Hints & Solutions

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$$\Rightarrow r^2 + 14\sqrt{5}r - 5 = 0$$

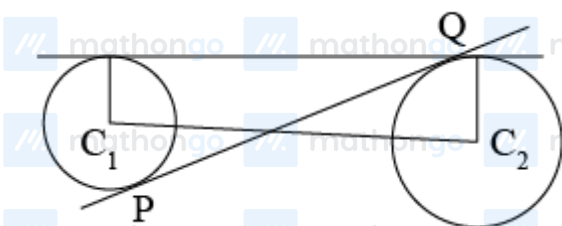
$$\therefore r = \sqrt{250} - \sqrt{245}$$

$$\text{Thus } \alpha - \beta = 5$$

Q40 (5) - Integer Type

The given equation of the circles are

$$x^2 + y^2 - 20x + 64 = 0$$



Here, $C_1(10, 0)$ and $r_1 = 6$

$$\text{and } x^2 + y^2 + 30x + 144 = 0$$

Here, $C_2(-15, 0)$ and $r_2 = 9$

$$|C_1 - C_2| = \sqrt{(10 + 15)^2 + (0 - 0)^2} = 25$$

$$r_1 + r_2 = 15$$

$$|C_1 C_2| = r_1 + r_2$$

So, there are two direct and two transverse common tangents. The length of the transverse common tangent is

less than the direct tangent. The length is given by

$$\begin{aligned} k &= \sqrt{(C_1 C_2)^2 - (r_1 + r_2)^2} \\ &= \sqrt{(25)^2 - (9 + 6)^2} = \sqrt{400} \end{aligned}$$

$$k = 20$$

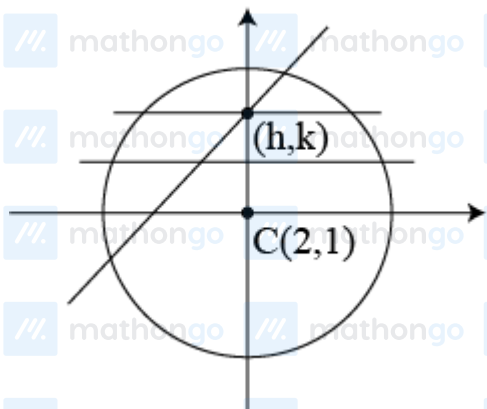
$$\text{So, value of } \frac{k}{4} = \frac{20}{4} = 5$$

Q41 (1) - Integer Type

Hints & Solutions

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Locus will be a line with slope $\left(-\frac{1}{2}\right)$ and passing through the centre $(2,1)$ of the circle $y - 1 = -\frac{1}{2}(x - 2)$



$$\Rightarrow x + 2y - 4 = 0$$

$$\therefore |a + b + c| = 1$$

Q42 (0) - Integer Type

$$P = a^2 + b^2 - 2(t^2 - 3t + 1)a - 2(t^2 + 2t)b + t$$

$$= -(2a + 2b)t^2 + (6a - 4b + 1)t + (a^2 + b^2 - 2a)$$

This power is independent of the parameter t . If and only if, $2a + 2b = 0 \Rightarrow a = -b$.

$$\text{and } 6a - 4b + 1 = 0$$

$$\therefore a = -\frac{1}{10} \text{ and } b = \frac{1}{10}$$

$$\Rightarrow a + b = 0$$

Q43 (5) - Integer Type

Let the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ as centre lie on $2x - 2y + 9 = 0$

$$\therefore -2g + 2f + 9 = 0 \dots(i)$$

as it cuts $x^2 + y^2 = 4$, orthogonally.

$$\therefore 2g(0) + 2f(0) = c - 4 \Rightarrow c = 4$$

Equation of circle

$$x^2 + y^2 + (2f + 9)x + 2fy + 4 = 0$$

$$\text{or } (x^2 + y^2 + 9x + 4) + 2f(x + y) = 0$$

It passes through point of intersection of

$$x^2 + y^2 + 9x + 4 = 0$$

$$\text{and } x + y = 0$$

Hints & Solutions

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$\therefore$ Points are $\left(-\frac{1}{2}, \frac{1}{2}\right)$ and $(-4, 4)$.

Hence, $a = \frac{-1}{2}, c = -4, b = \frac{1}{2}, d = 4$

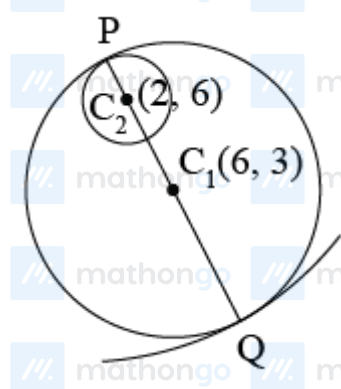
Thus, $2b + d = 5$

Q44 (4) - Integer Type

First equation is

$$(x - 6)^2 + (y - 3)^2 = 49 \dots (i)$$

$$(x - 2)^2 + (y - 6)^2 = 40 + k \dots (ii)$$



C_1 : centre $(6, 3)$ and $r_1 = 7$

C_2 : centre $(2, 6); r_2 = \sqrt{40 + k}$

The two circles will intersect is

$$C_2P \geq 7 - 5$$

$$\Rightarrow \sqrt{40 + k} \geq 2$$

$$\Rightarrow k \geq -36 \dots (iii)$$

and $C_2Q \leq 7 + 5$

$$\Rightarrow 40 + k \leq 12^2 \dots (iv)$$

$$\Rightarrow k \leq 104$$

$$\therefore -36 \leq k \leq 104$$

$$\Rightarrow a = -36$$

$$b = 104$$

Thus, $|3a + b| = 4$

Q45 (5) - Integer Type

Hints & Solutions

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Given equations of circle C_1 is

$$x^2 + y^2 - 20x + 64 = 0$$

and C_2

$$x^2 + y^2 + 30x + 144 = 0$$

Centre and radius of C_1 are (10,0) and 6 and C_2 are (-15,0) and 9, respectively.

PQ will be indirect tangent for circles C_1 and C_2

$$\text{So, length of } PQ, \text{ i.e. } l = \sqrt{d^2 - (r_1 + r_2)^2}$$

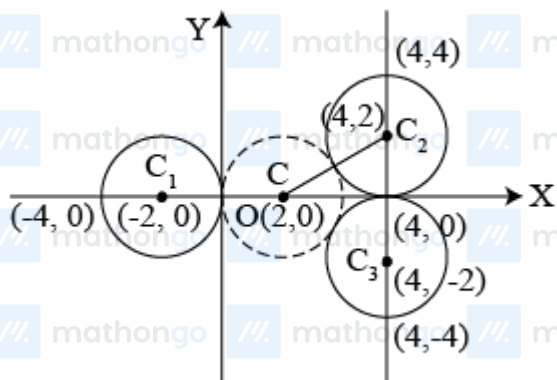
where, d is distance between two circles and r_1 and r_2 are their radi Now,

$$l = \sqrt{625 - (9 + 6)^2} = \sqrt{625 - 225} = \sqrt{400} = 20$$

So, the value of $(l - 15)$ is 5 .

Q46 (7) - Integer Type

$$CC_1 = CC_2 = CC_3 = r + 2$$



$$\Rightarrow (r + 2)^2 = (r - 4)^2 + 4 = (r - 4)^2 + 4$$

$$\Rightarrow r^2 + 4r + 4 = r^2 - 8r + 20$$

$$\Rightarrow 12r = 16 \text{ or } r = \frac{4}{3} = \frac{p}{q}$$

$$\Rightarrow p + q = 7$$

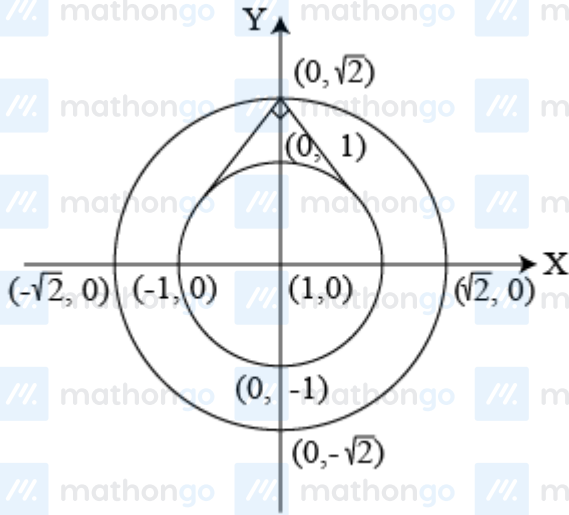
Q47 (2) - Integer Type

The intersection point of tangents lie on $x^2 + y^2 = 2$ (director's circle)

$\therefore$ Maximum value of square of ordinate of the point from where tangents are drawn is $(\sqrt{2})^2 = 2$

Hints & Solutions

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Q48 (3) - Integer Type

Let $\sum_{i=1}^6 x_i = \alpha$ and $\sum_{i=1}^6 y_i = \beta$

Let O be the orthocentre of the triangle made by (x_1, y_1) , (x_2, y_2) and (x_3, y_3)

$$\Rightarrow O \text{ is } (x_1 + x_2 + x_3, y_1 + y_2 + y_3) = (\alpha_1, \beta_1)$$

Similarly, let G be the centroid of the triangle made by other three points,

$$\Rightarrow G \left(\frac{x_4 + x_5 + x_6}{3}, \frac{y_4 + y_5 + y_6}{3} \right)$$

$$\Rightarrow G \left(\frac{\alpha - \alpha_1}{3}, \frac{\beta - \beta_1}{3} \right)$$

The point dividing OG in the ratio 3: 1 is $\left(\frac{\alpha}{4}, \frac{\beta}{4} \right) = (2, 1)$

$$\Rightarrow h + k = 2 + 1 = 3$$

Q49 (9) - Integer Type

$$f(x, y) = x^2 + y^2 + 2gx + 2fy + c = 0$$

$$\text{Now, } f(0, \lambda) = \lambda^2 + 2f\lambda + c = 0$$

and its roots are $(1, 1)$

$$\Rightarrow f = -1, c = 1$$

$$f(\lambda, 0) = \lambda^2 + 2g\lambda + c = 0 \text{ and its roots are } \left(\frac{1}{5}, 5 \right)$$

Hints & Solutions

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$$\therefore \frac{1}{5} + 5 = -2g$$

$$\Rightarrow g = \frac{-13}{5}$$

$$\text{and } \frac{1}{5} \times 5 = c$$

$$\text{Thus, } g = \frac{-13}{5}$$

$$\Rightarrow f = -1, c = 1$$

$$r = \sqrt{\frac{169}{25} + 1} - 1 = \frac{13}{5}$$

$$\therefore 5r - 4 = 9$$

Q50 (1) - Integer Type

Let the point $\left(m_i, \frac{1}{m_i}\right), i = 1, 2, 3, 4$

lie on the circle $x^2 + y^2 + 2gx + 2fy + c = 0$.

$$\text{Then, } m_i^2 + \frac{1}{m_i^2} + 2gm_i + \frac{2f}{m_i} + c = 0, i = 1, 2, 3, 4$$

$$\Rightarrow m_i^4 + 2gm_i^3 + cm_i^2 + 2fm_i + 1 = 0, i = 1, 2, 3, 4$$

m_1, m_2, m_3 and m_4 are roots of the equation.

$$m^4 + 2gm^3 + cm^2 + 2fm + 1 = 0$$

$$\Rightarrow m_1m_2m_3m_4 = \frac{1}{1} = 1$$